POWER IN - USB-C 5V, TVS D3, PTC F1, ferrite FB1

J1
USB-C 16P (USB4105-GF-A)
VBUS A4
CC1 A5
CC2 B5
D- A7 USB_DM
D- B7 USB_DM
D+ A6 USB_DP
D+ B6 USB_DP
SBU1 A8
SBU2 B8
SHIELD
GND A1
SH
CC1
CC2
R3 5k1
R4 5k1
D3 SMF5.0A (TVS)
DNP
VBUS
GND
PWR_FLAG
VBUS_F
F1 PTC 0.5A (MF-NSMF050-2)
C2 100n
10u
FB1 600R@100MHz
+5V

HV BOOST 75-100 V (MCU-controlled, TIM2_CH3 on PD6, 30 kHz)

L11 330u (SWPA6028S331MT, 200V-1.4Ω)
D1 SOD-123F (verify P/N)
SW
C3 22u 10V
R1 47
GATE_DRV
GATE
Q1 DMN15H310SE
R2 100k
R12 100 (snubber)
C8 100p C0G 250V
C4 1u 250V
D2 SMF100A
R5 1M
R6 1M
R7 39k
C5 10n
HV+
VFB

Q1 DMN15H310SE: 150 V, 0.33 Ω @ Vgs=5 V. R2 keeps boost off during reset.
D2 = hard OVP clamp (Vbr 111-123 V). R12/C8 snubber: optional (DNP).

MCU CH32V003F4P6 (runs at 5 V)

PD7 = SR_DATA: program NRST as GPIO (minichlink -D)

U1 CH32V003F4P6
PC0 LAMP7
PC1 LAMP4
PC2 LAMP2
PC3 LAMP5
PC4 LAMP3
PC5 LAMP9
PC6 LAMP6
PC7 SR_CLK
PD0 LAMP8
PD1 SWIO
PD2 USB_DP_MCU
PD3 VFB
PD4 USB_DM_MCU
PD5 SR_LATCH
PD6 GATE_DRV
PD7 SR_DATA
PA1 LDR
PA2 TOUCH2
VDD
VSS
C6 100n

J2 SWD pogo pads
SWIO
GND

INPUTS - LDR, TKEY touch pads, spare pad

Q2 TEPT4400
LDR
R9 22k
TP2 Touch trunk
TOUCH2
R11 1k
TP3 SR_CLK test
SR_CLK

USB low-speed (rv003usb): D+ = PD2, D- = PD4.
The MCU stays at 5 V: 68R + 3.6 V zeners clamp its drive to USB levels. 1k5 D- pull-up to VBUS.

R15 1k5
R13 68
R14 68
USB_DP
USB_DM
VBUS
D4 BZX884-C3V6
D5 BZX884-C3V6

LAMP DRIVE - 2 × 74HC595: LAMPA1-10 (anode-side switches) + the cathode switches of LA1 and LA10

SR_DATA = PD7, SR_CLK = PC7, SR_LATCH = PD5. /OE tied low: outputs are random until the first latch, but the boost is off then (R2).

U2 74HC595BQ
SER SR_DATA
SRCLK SR_CLK
SRCLR +5V
RCLK SR_LATCH
OE GND
QA LAMPA8
QB LAMPA7
QC LAMPA4
QD LAMPA2
QE LAMPA1
QF LAMPA5
QG LAMPA3
QH SR_CHAIN
QH'
VCC
GND
C9 100n

U3 74HC595BQ
SER SR_CHAIN
SRCLK SR_CLK
SRCLR +5V
RCLK SR_LATCH
OE GND
QA
QB
QC LAMPA10
QD LAMPA9
QE LAMPA6
QF
QG
QH
QH'
VCC
GND
C10 100n

NEON LAMPS - Brightmaster NEON-3 (strike ≤90 VDC, ~60 V settled). Each lead: 150k (200 V type) to HV+ and one half of an MMDT5551 (160 V) to GND. Pull one lead low: that electrode glows.

I_lamp = (V_HV − 60 V)/150k ≈ 0.1-0.3 mA, plus V_HV/150k ≈ 0.7 mA lost in the pulled-low lead's own pull-up. Both off: dark, no current.

HV+
LA1 150k 200V NEON-3
LA2 150k 200V NEON-3
LA3 150k 200V NEON-3
LA4 150k 200V NEON-3
LA5 150k 200V NEON-3
LA6 150k 200V NEON-3
LA7 150k 200V NEON-3
LA8 150k 200V NEON-3
LA9 150k 200V NEON-3
LA10 150k 200V NEON-3
R21 R31 R22 R32 R23 R33 R24 R34 R25 R35
R26 R36 R27 R37 R28 R38 R29 R39 R30 R40
R51 R41 R52 R42 R53 R43 R54 R44 R55 R45
R56 R46 R57 R47 R58 R48 R59 R49 R60 R50
22k
Q11B Q11A Q12B Q12A Q13B Q13A Q14B Q14A Q15B Q15A
Q16B Q16A Q17B Q17A Q18B Q18A Q19B Q19A Q20B Q20A
MMDT5551
LAMPA1 LAMP1 LAMPA2 LAMP2 LAMPA3 LAMP3 LAMPA4 LAMP4 LAMPA5 LAMP5
LAMPA6 LAMP6 LAMPA7 LAMP7 LAMPA8 LAMP8 LAMPA9 LAMP9 LAMPA10 LAMP10
GND

Sheet: /
File: NeonPine.kicad_sch
Title: NeonPine - neon Christmas tree night light
Size: A3 | Date: 2026-10-05 | **Rev: B**
KiCad E.D.A. 10.0.6 | Id: 1/1